

Entropy-Rate Bounds for Stochastic Codewords

Typical-Set Packing and the Binary Symmetric Channel Example

Technical Note

September 2026

Abstract

A stochastic codeword is a probability law over length- L output strings rather than a single deterministic string. If each codeword law is generated by a stationary ergodic binary source of entropy rate h , the asymptotic equipartition property (AEP) says that essentially all of its probability is carried by about 2^{Lh} strings. Since the ambient binary space contains 2^L strings, a simple packing argument suggests at most $2^{L(1-h)}$ reliably distinguishable stochastic codewords, or rate $1 - h$ bits per emitted symbol. This note states a precise version of that converse under a uniform AEP condition, separates the converse from the additional assumptions needed for achievability, and gives the binary symmetric channel (BSC) as a classical construction in which the corresponding stochastic-cloud rate $1 - H_2(p)$ is actually achievable. The BSC example also shows why low entropy is essential for rates close to one. Finally, we state the open PFSA question: whether a sufficiently large family of low-entropy dynamical sources can attain the same packing exponent while remaining identifiable from finite realizations.

1 Stochastic codewords

Let $\mathcal{X} = \{0, 1\}$. For block length L , a stochastic codebook contains M_L probability laws

$$P_{m,L} \quad (m = 1, \dots, M_L)$$

on $\mathcal{X}^L$. To transmit message m , the encoder draws

$$X^L \sim P_{m,L}.$$

A decoder $\phi_L : \mathcal{X}^L \rightarrow \{1, \dots, M_L\}$ estimates the message. The maximal error probability is

$$P_e^{(L)} = \max_m P_{m,L} \{\phi_L(X^L) \neq m\}.$$

The code rate is

$$R_L = \frac{1}{L} \log_2 M_L \quad \text{bits per emitted binary symbol.}$$

The case of interest is that each $P_{m,L}$ is the length- L marginal of a stationary ergodic stochastic source G_m with entropy rate $h(G_m) = h$. A message is therefore represented by the *identity of a stochastic generator*; the actual observed string is one random realization from that generator.

2 AEP volume of one stochastic codeword

For a fixed stationary ergodic source G with entropy rate h , the Shannon–McMillan–Breiman theorem gives

$$-\frac{1}{L} \log_2 P_G(X^L) \longrightarrow h$$

almost surely. Equivalently, for every $\epsilon > 0$, there are typical sets $\mathcal{T}_{G,L}^{(\epsilon)}$ such that

$$P_G(\mathcal{T}_{G,L}^{(\epsilon)}) \rightarrow 1$$

and every $x^L \in \mathcal{T}_{G,L}^{(\epsilon)}$ satisfies

$$2^{-L(h+\epsilon)} \leq P_G(x^L) \leq 2^{-L(h-\epsilon)}.$$

Consequently,

$$2^{L(h-\epsilon)}(1 - o(1)) \leq |\mathcal{T}_{G,L}^{(\epsilon)}| \leq 2^{L(h+\epsilon)}.$$

Thus, at exponential scale, one entropy-rate- h stochastic codeword occupies 2^{Lh} ordinary binary strings.

This is the origin of the heuristic

$$\frac{2^L}{2^{Lh}} = 2^{L(1-h)}$$

for the number of stochastic codeword clouds that might fit in the ambient sequence space. To make this a converse for a growing family of sources, however, one needs a uniform form of the AEP.

3 A precise rate converse

Definition 1 (Uniform AEP family). *A sequence of stochastic codebooks $\{P_{m,L} : 1 \leq m \leq M_L\}$ has entropy exponent h uniformly if for every $\epsilon > 0$ there exists L_0 such that, for all $L \geq L_0$ and every $m \leq M_L$, there is a set $\mathcal{T}_{m,L}$ satisfying*

$$P_{m,L}(\mathcal{T}_{m,L}) \geq 1 - \epsilon$$

and

$$P_{m,L}(x^L) \leq 2^{-L(h-\epsilon)} \quad \text{for every } x^L \in \mathcal{T}_{m,L}.$$

For a single fixed stationary ergodic source this follows from AEP; for a codebook whose number of distinct sources grows exponentially with L , uniformity is an additional condition and must not be silently assumed.

Theorem 1 (Entropy-rate packing converse). *Consider binary stochastic codebooks satisfying the uniform AEP condition with entropy exponent h . If the maximal decoding error obeys $P_e^{(L)} \rightarrow 0$, then*

$$\limsup_{L \rightarrow \infty} \frac{1}{L} \log_2 M_L \leq 1 - h.$$

Equivalently, every reliably decodable rate satisfies

$$\boxed{R \leq 1 - h.}$$

Proof. Let

$$D_{m,L} = \{x^L : \phi_L(x^L) = m\}$$

be the decoder region for message m . The $D_{m,L}$ are pairwise disjoint and, by maximal reliability,

$$P_{m,L}(D_{m,L}) \geq 1 - P_e^{(L)}.$$

Fix $\epsilon > 0$ and choose the uniform typical set $\mathcal{T}_{m,L}$. Then

$$P_{m,L}(D_{m,L} \cap \mathcal{T}_{m,L}) \geq 1 - P_e^{(L)} - \epsilon.$$

Every point of $\mathcal{T}_{m,L}$ has probability at most $2^{-L(h-\epsilon)}$. Therefore

$$1 - P_e^{(L)} - \epsilon \leq |D_{m,L} \cap \mathcal{T}_{m,L}| 2^{-L(h-\epsilon)},$$

so

$$|D_{m,L}| \geq (1 - P_e^{(L)} - \epsilon) 2^{L(h-\epsilon)}.$$

Summing over messages and using disjointness,

$$\begin{aligned} 2^L &\geq \sum_{m=1}^{M_L} |D_{m,L}| \\ &\geq M_L (1 - P_e^{(L)} - \epsilon) 2^{L(h-\epsilon)}. \end{aligned}$$

Hence

$$\frac{1}{L} \log_2 M_L \leq 1 - h + \epsilon - \frac{1}{L} \log_2 (1 - P_e^{(L)} - \epsilon).$$

Take $L \rightarrow \infty$, then $\epsilon \downarrow 0$, to obtain

$$\limsup_{L \rightarrow \infty} R_L \leq 1 - h.$$

□

Remark 1 (What the theorem does and does not prove). *The theorem is a converse. AEP determines the exponential volume required by each stochastic codeword, but it does not itself prove that the ambient space can be tiled or packed by $2^{L(1-h)}$ distinguishable stationary ergodic sources. Equality $R = 1 - h$ requires an achievability construction with sufficiently separated probability laws.*

4 Conditional achievability statement

The exact missing ingredient can be stated cleanly.

Proposition 1 (Separated-family achievability). *Suppose that for every $R < 1 - h$ and all sufficiently large L there exist*

$$M_L = \lfloor 2^{LR} \rfloor$$

stochastic laws $P_{m,L}$ and pairwise disjoint sets $D_{m,L} \subseteq \mathcal{X}^L$ such that

$$\inf_{m \leq M_L} P_{m,L}(D_{m,L}) \longrightarrow 1.$$

Then every $R < 1 - h$ is achievable with vanishing maximal error. Combined with the preceding converse, the stochastic-code capacity of that admissible family is $1 - h$.

Proof. Use the decoder $\phi_L(x^L) = m$ when $x^L \in D_{m,L}$, with an arbitrary rule outside their union. Since the $D_{m,L}$ are disjoint,

$$P_e^{(L)} \leq 1 - \inf_m P_{m,L}(D_{m,L}) \rightarrow 0.$$

Thus each $R < 1 - h$ is achievable. The converse prevents any $R > 1 - h$ under the same uniform entropy condition. $\square$

The proposition deliberately isolates the hard part: constructing exponentially many *different stochastic generators* whose high-probability regions remain distinguishable.

5 Why low entropy gives high rate

The binary ambient space has exponent 1 because $|\mathcal{X}^L| = 2^L$. A source with entropy rate h consumes exponent h . The remaining exponent available for source identity is therefore at most

$$1 - h.$$

Thus low stochastic entropy is not incidental; it is the mechanism by which the identity of the generating process can carry a high rate.

At the extremes:

- If $h \rightarrow 0$, one stochastic codeword occupies only $2^{o(L)}$ strings, so the counting bound approaches one bit per emitted binary symbol.
- If $h \rightarrow 1$, one source occupies essentially all 2^L binary strings at exponential scale, leaving zero exponential room for source identity.

For an alphabet of size q , the identical argument gives

$$R \leq \log_2 q - h,$$

and the normalized rate is at most

$$1 - \frac{h}{\log_2 q}.$$

6 The binary symmetric channel as an achievable stochastic-cloud example

The classical BSC gives a concrete example where the counting exponent is not merely an upper bound but the exact achievable rate.

Choose a deterministic binary center $c_m \in \{0, 1\}^L$ for message m and transmit

$$X^L = c_m \oplus Z^L, \quad Z_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p), \quad 0 < p < \frac{1}{2}.$$

Conditioned on c_m , the transmitted string is therefore a stochastic cloud around c_m . The noise entropy per symbol is

$$h = H_2(p) = -p \log_2 p - (1 - p) \log_2 (1 - p).$$

With high probability, Z^L has approximately pL ones. The number of such typical noise patterns is

$$\binom{L}{pL} \doteq 2^{LH_2(p)},$$

where $a_L \doteq 2^{L\alpha}$ denotes equality at exponential order. Consequently the typical received cloud around one codeword has volume exponent $H_2(p)$, while the whole binary space has exponent 1. The packing heuristic gives

$$M_L \doteq 2^{L[1-H_2(p)]}.$$

For the BSC, Shannon’s channel coding theorem proves achievability: for every

$$R < 1 - H_2(p)$$

there exist length- L codebooks with $M_L \approx 2^{LR}$ and decoding error tending to zero. Conversely, no larger rate is reliable. Hence

$$C_{\text{BSC}(p)} = 1 - H_2(p).$$

This is exactly the same exponent obtained by “ambient volume minus stochastic-cloud volume.” The BSC therefore provides a rigorous existence example of high-rate stochastic-cloud coding.

6.1 Numerical examples

For $p = 0.01$,

$$H_2(0.01) \approx 0.080793, \quad C \approx 0.919207 \text{ bits/symbol.}$$

Thus a stochastic cloud containing approximately $2^{0.0808L}$ typical realizations per deterministic center still permits roughly $2^{0.9192L}$ reliably distinguishable messages.

For $p = 0.001$,

$$H_2(0.001) \approx 0.011408, \quad C \approx 0.988592 \text{ bits/symbol.}$$

Only one tenth of one percent crossover probability already gives a genuinely stochastic transmitted sequence while allowing a rate within about 1.14% of the binary maximum.

p	$H_2(p)$	$1 - H_2(p)$
0.10	0.468996	0.531004
0.05	0.286397	0.713603
0.01	0.080793	0.919207
0.001	0.011408	0.988592

7 Relation to PFSA stochastic coding

The BSC example should be interpreted carefully. Its message-dependent law is a noisy cloud centered on a deterministic length- L codeword. It proves that the typical-set packing exponent $1 - h$ can be achievable for stochastic codeword clouds. It does *not* by itself prove that an exponentially large family of stationary finite-state stochastic processes can attain the same exponent.

For PFSA coding, the desired construction is different. A message m selects a PFSA G_m , and the observed codeword is a length- L realization

$$X^L \sim G_m.$$

The decoder must infer which dynamical law generated the realization. If all machines have entropy rate approximately h , the converse above suggests the same fundamental ceiling $1 - h$, but achievability requires enough process diversity and enough statistical separation.

A precise target theorem would therefore have the following form.

For every sufficiently small h and every $R < 1 - h$, construct a family of $M_L \geq 2^{LR}$ finite-stationary ergodic processes of entropy rate $h + o(1)$ such that the generating process can be identified from a length- L realization with maximal error probability tending to zero.

Proving this would establish

$$C_{\text{PFSA}}(h) = 1 - h$$

for the specified admissible PFSA class. Without such a construction, $1 - h$ should be called the entropy-volume converse or packing limit, not yet the PFSA capacity.

8 Connection to delimiter-free process coding

The recent delimiter-free experiment illustrates a complementary issue. There, information was represented by latent process identity and residence duration. A decoder recovered process changes from higher-order temporal structure without being supplied explicit delimiters. That experiment demonstrates finite-sample *identifiability* of stochastic regimes; the theorem here concerns the asymptotic *number* of distinct stochastic codewords that can coexist at a given entropy rate.

These are the two ingredients needed for a full stochastic coding theory:

1. **Packing:** how many stochastic generators can be placed in sequence space at entropy rate h ?
2. **Identification:** how long a realization is required to determine the active generator with a specified error probability?

The AEP gives the natural packing exponent $1 - h$; process divergence and finite-sample discrimination determine whether that exponent can actually be attained by a structured family such as PFSA.

9 Summary

For binary stochastic codewords with a uniform entropy exponent h , reliable decoding implies

$$R \leq 1 - h.$$

The proof is a direct consequence of AEP volume and disjoint decoding regions. Equality is not automatic: it requires a family whose high-probability regions can be packed and distinguished at the corresponding exponential rate. The BSC supplies the canonical achievable example,

$$C = 1 - H_2(p),$$

showing explicitly that low stochastic entropy permits rates arbitrarily close to one. The central PFSA problem is therefore an achievability problem: determine whether low-entropy dynamical generators can be constructed in exponentially large, reliably distinguishable families approaching the same $1 - h$ exponent.